CS 527 – Database Systems Project Requirements Specification for the Programming Project

Introduction

In this project, you will design and implement a relational database system to support the operations of an **online auction system**. You will use HTML for the user interface and MySQL for the database server. You will have to install your own web server to host your web application, as well as a MySQL server**. Everything will be local.**

You are to work in **teams of four**.

Project Specification

As you probably know, there are a multitude of online auction systems on the web. The most popular one is eBay. I suggest you visit this website (although I suppose all of you have visited it many times) to understand the look-and-feel of an auction website and how such a system is supposed to function.

The basic idea behind your online auction system is to support a company that lets users buy and sell items in an online auction. (Think *eBay*. So we will call the company that hired you, BuyMe. If you are not familiar with how *eBay* works, please go to eBay to look for something to buy and review its FAQ/Help pages.)

In general, a seller posts an item for sale, starting an auction that will close at a specified date and time. The seller also posts an initial price, an increment for bids [which indicates a lower bound on how much must be added to the current bid for the next valid bid], and a (secret) minimum price [the seller is not willing to accept less than that]. Potential buyers post bids during the auction, and the user with the highest bid at closing time wins the item. (Actually, they *must* buy the item.) The bidding process is described in the Help pages above. The one fancy thing you need to implement is "automatic bidding", which involves the buyer setting a (secret) upper limit on how much they are willing to pay; every time someone bids higher than your current bid, the computer automatically puts in a higher bid for you, till your upper limit is reached. So if there is not much bidding, you may get the item for less than your upper limit. If someone bids past your upper limit, BuyMe will alert you.
NOTE: Each team's BuyMe site will be restricted to a specific category of items (e.g., vehicles OR clothing OR computers ...) with at least three hierarchical subcategories (eg, truck, motorbike, car, foreignCar, for Vehicles), which **require** certain specific fields to be filled for those kinds of items. (*motivation*: I have been annoyed on eBay when people do not mention shoe size or belt width and length.) The choice of category is up to each group in the course. [**Bonus**

points for any team that can *change the kinds of items available* without recompiling the code (i.e., be driven by data in the database).] Additional features for your BuyMe (some of which are not available on eBay) depend on the three classes of users supported:

End-users (buyers and sellers)

1. They must, of course, be able to create and delete accounts, and log in and log out. 2. An end-user can search the list of items on auction according to various criteria based on the fields describing an item. (The more complex searches your team supports, the higher the credit.)
3. Potential buyers should be able to set **alerts** for items they are interested in buying. (*motivation*: I am often looking for certain items (e.g., sandals, winter coats) that are no longer manufactured. I am interested when such items show up, esp. if I could also specify sizes & colors.)
4. A user should be able to view the *history of bids* for any specific auction, the list of all auctions a specific buyer or seller has participated in, and the list of "similar" items on auction in the preceding month (and auction information about them). [You get to decide the "similarity measure".]
5. If you wish, you can anonymize end-users so their actual login-names and e-mail addresses do not show up in auctions.

In addition to end users, the BuyMe system must support its staff.

II. Customer representatives are available to end-users for answering questions and modifying any information, as long as the customer rep decides this is reasonable. (This includes resetting passwords and removing bids. So your system need not support any specific rules for removing bids.) They must therefore be able to perform such actions and remove illegal auctions.

III. **One** *administrative staff member*, whose account will have been created ahead of time, will be able to create accounts for customer representatives. This person must also be able to generate summary sales reports, including: total earnings; earnings per {item, item type, end-user}; and best-selling {items, end-users}.

SQLAlchemy + Flask sample app: https://github.com/zhetang1/cs527/tree/master

Please create a private GitHub repo, invite pranikamassey@gmail.com and zhetang1@gmail.com. Please commit all your files, including the ER diagrams. Please let team members know in the README.

Please use GitHub to collaborate. I expect all team members to commit code, not just one student, to commit all changes. Only Code commits are considered contributions. If you didn't commit code, you are considered not to have contributed to the project.

Feature List

Create user accounts and log in and out.

I. Auctions

[] seller creates auctions and posts items for sale

[] set all the characteristics of the item

[] set closing date and time

[] set a hidden minimum price (reserve)

[] A buyer should be able to bid

[] let the buyer set a new bid

[] in case of automatic bidding, set a secret upper limit and bid increment [] alert other buyers of the item that a higher bid has been placed (manual) [] alert buyers in case someone bids more than their upper limit (automatic) [] define the winner of the auction

[] When the closing time has come, check if the seller has set a reserve. [] if yes: if the reserve is higher than the last bid, none is the winner. [] if no: whoever has the higher bid is the winner

[] alert the winner that they won the auction

II. Browsing and advanced search functionality

[] let people browse the items and see the status of the current bidding [] sorted by different criteria (by type, bidding price, etc.)

[] Search the list of items by various criteria.

[] a user should be able to:

[] view all the history of bids for any specific auction

[] view the list of all auctions a specific buyer or seller has participated in [] view the list of "similar" items on auctions in the preceding month (and auction information about them)

[] let the user set an alert for specific items s/he is interested in

[] get an alert when the item becomes available

[] users browse questions and answers

[] users search questions by keywords

I||. Admin and customer rep functions

[] Admin (create an admin account ahead of time)

[] creates accounts for customer representatives

[] generates sales reports for:

[] total earnings

[] earnings per: item, item type, end-user, etc.

[] best-selling items

[] best buyers

[] Customer representative functions:

[] users post questions to the customer representatives (i.e., customer service) [] reps reply to user questions

[] edits and deletes account information

[] removes bids

[] removes auctions